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| Tool Version : Vivado v.2023.1 (win64) Build 3865809 Sun May  7 15:05:29 MDT 2023  
| Date          : Wed Sep  9 13:14:33 2026  
| Host          : bedouinoureddine running 64-bit major release (build 9200)  
| Command       : report_utilization -file  
C:/Vivado_Projects/Harness_Test_System_3/Harness_Test_System/utilization_report2.tx  
t -name utilization_2  
| Design        : FPGA_Top  
| Device        : xc7a100tcsg324-1  
| Speed File    : -1  
| Design State  : Physopt postRoute  
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```

Utilization Design Information

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### 1. Slice Logic

| Site Type             | Used | Fixed | Prohibited | Available | Util% |
|-----------------------|------|-------|------------|-----------|-------|
| Slice LUTs            | 744  | 0     | 0          | 63400     | 1.17  |
| LUT as Logic          | 744  | 0     | 0          | 63400     | 1.17  |
| LUT as Memory         | 0    | 0     | 0          | 19000     | 0.00  |
| Slice Registers       | 1039 | 0     | 0          | 126800    | 0.82  |
| Register as Flip Flop | 1039 | 0     | 0          | 126800    | 0.82  |
| Register as Latch     | 0    | 0     | 0          | 126800    | 0.00  |
| F7 Muxes              | 0    | 0     | 0          | 31700     | 0.00  |
| F8 Muxes              | 0    | 0     | 0          | 15850     | 0.00  |

\* Warning! LUT value is adjusted to account for LUT combining.

## 1.1 Summary of Registers by Type

| Total | Clock Enable | Synchronous | Asynchronous |
|-------|--------------|-------------|--------------|
| 0     | -            | -           | -            |
| 0     | -            | -           | Set          |
| 0     | -            | -           | Reset        |
| 0     | -            | Set         | -            |
| 0     | -            | Reset       | -            |
| 0     | Yes          | -           | -            |
| 0     | Yes          | -           | Set          |
| 1039  | Yes          | -           | Reset        |
| 0     | Yes          | Set         | -            |
| 0     | Yes          | Reset       | -            |

## 2. Slice Logic Distribution

| Available | Util% | Site Type              | Used | Fixed | Prohibited |
|-----------|-------|------------------------|------|-------|------------|
| 15850     | 2.36  | Slice                  | 374  | 0     | 0          |
|           |       | SLICEL                 | 242  | 0     |            |
|           |       | SLICEM                 | 132  | 0     |            |
| 63400     | 1.17  | LUT as Logic           | 744  | 0     | 0          |
|           |       | using 05 output only   | 6    |       |            |
|           |       | using 06 output only   | 227  |       |            |
|           |       | using 05 and 06        | 511  |       |            |
| 19000     | 0.00  | LUT as Memory          | 0    | 0     | 0          |
|           |       | LUT as Distributed RAM | 0    | 0     |            |
|           |       | LUT as Shift Register  | 0    | 0     |            |
|           |       | Slice Registers        | 1039 | 0     | 0          |

|                                        |      |  |  |   |  |
|----------------------------------------|------|--|--|---|--|
| 126800                                 | 0.82 |  |  |   |  |
| Register driven from within the Slice  | 53   |  |  |   |  |
| Register driven from outside the Slice | 986  |  |  |   |  |
| LUT in front of the register is unused | 871  |  |  |   |  |
| LUT in front of the register is used   | 115  |  |  |   |  |
| Unique Control Sets                    | 2    |  |  | 0 |  |
| 15850                                  | 0.01 |  |  |   |  |

\* \* Note: Available Control Sets calculated as Slice \* 1, Review the Control Sets Report for more information regarding control sets.

### 3. Memory

| Site Type      | Used | Fixed | Prohibited | Available | Util% |
|----------------|------|-------|------------|-----------|-------|
| Block RAM Tile | 0    | 0     | 0          | 135       | 0.00  |
| RAMB36/FIFO*   | 0    | 0     | 0          | 135       | 0.00  |
| RAMB18         | 0    | 0     | 0          | 270       | 0.00  |

\* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

### 4. DSP

| Site Type    | Used | Fixed | Prohibited | Available | Util% |
|--------------|------|-------|------------|-----------|-------|
| DSPs         | 2    | 0     | 0          | 240       | 0.83  |
| DSP48E1 only | 2    |       |            |           |       |

### 5. IO and GT Specific

| Site Type  | Used | Fixed | Prohibited | Available | Util% |
|------------|------|-------|------------|-----------|-------|
| Bonded IOB | 20   | 20    | 0          | 210       | 9.52  |

|                             |    |   |   |     |      |  |
|-----------------------------|----|---|---|-----|------|--|
| IOB Master Pads             | 9  |   |   |     |      |  |
| IOB Slave Pads              | 10 |   |   |     |      |  |
| Bonded IPADs                | 0  | 0 | 0 | 2   | 0.00 |  |
| PHY_CONTROL                 | 0  | 0 | 0 | 6   | 0.00 |  |
| PHASER_REF                  | 0  | 0 | 0 | 6   | 0.00 |  |
| OUT_FIFO                    | 0  | 0 | 0 | 24  | 0.00 |  |
| IN_FIFO                     | 0  | 0 | 0 | 24  | 0.00 |  |
| IDELAYCTRL                  | 0  | 0 | 0 | 6   | 0.00 |  |
| IBUFDS                      | 0  | 0 | 0 | 202 | 0.00 |  |
| PHASER_OUT/PHASER_OUT_PHY   | 0  | 0 | 0 | 24  | 0.00 |  |
| PHASER_IN/PHASER_IN_PHY     | 0  | 0 | 0 | 24  | 0.00 |  |
| IDELAYE2/IDELAYE2_FINEDELAY | 0  | 0 | 0 | 300 | 0.00 |  |
| ILOGIC                      | 0  | 0 | 0 | 210 | 0.00 |  |
| OLOGIC                      | 0  | 0 | 0 | 210 | 0.00 |  |

## 6. Clocking

| Site Type  | Used | Fixed | Prohibited | Available | Util% |
|------------|------|-------|------------|-----------|-------|
| BUFGCTRL   | 2    | 0     | 0          | 32        | 6.25  |
| BUFIO      | 0    | 0     | 0          | 24        | 0.00  |
| MMCME2_ADV | 1    | 0     | 0          | 6         | 16.67 |
| PLLE2_ADV  | 0    | 0     | 0          | 6         | 0.00  |
| BUFMRCE    | 0    | 0     | 0          | 12        | 0.00  |
| BUFHCE     | 0    | 0     | 0          | 96        | 0.00  |
| BUFR       | 0    | 0     | 0          | 24        | 0.00  |

## 7. Specific Feature

| Site Type   | Used | Fixed | Prohibited | Available | Util% |
|-------------|------|-------|------------|-----------|-------|
| BSCANE2     | 0    | 0     | 0          | 4         | 0.00  |
| CAPTUREE2   | 0    | 0     | 0          | 1         | 0.00  |
| DNA_PORT    | 0    | 0     | 0          | 1         | 0.00  |
| EFUSE_USR   | 0    | 0     | 0          | 1         | 0.00  |
| FRAME_ECCE2 | 0    | 0     | 0          | 1         | 0.00  |
| ICAPE2      | 0    | 0     | 0          | 2         | 0.00  |
| PCIE_2_1    | 0    | 0     | 0          | 1         | 0.00  |
| STARTUPE2   | 0    | 0     | 0          | 1         | 0.00  |
| XADC        | 0    | 0     | 0          | 1         | 0.00  |

## 8. Primitives

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| Ref Name   | Used | Functional Category |
|------------|------|---------------------|
| FDCE       | 1039 | Flop & Latch        |
| LUT3       | 595  | LUT                 |
| LUT4       | 494  | LUT                 |
| CARRY4     | 181  | CarryLogic          |
| LUT2       | 63   | LUT                 |
| LUT6       | 53   | LUT                 |
| LUT5       | 27   | LUT                 |
| LUT1       | 23   | LUT                 |
| IBUF       | 18   | IO                  |
| OBUF       | 2    | IO                  |
| DSP48E1    | 2    | Block Arithmetic    |
| BUFG       | 2    | Clock               |
| MMCME2_ADV | 1    | Clock               |

## 9. Black Boxes

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| Ref Name | Used |
|----------|------|
|----------|------|

## 10. Instantiated Netlists

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| Ref Name  | Used |
|-----------|------|
| clk_wiz_0 | 1    |